

Civics Group		Name (use BLOCK LETTERS)
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H2

Centre number / Index Number	
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**ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIMS**

H2 BIOLOGY
9744/4
Paper 4: Practical Exam (Mark Scheme)

Friday

24th August 2018

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use a HB pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

IMPORTANT INFORMATION TO CANDIDATES:

Candidates with access to microscope at the start of the paper are given the **first 1h 15 min** to use them. Please answer **QUESTION 2** within this time frame.

Candidates with no access to microscope at the start of the paper will be given access **1h 15min after the start of the paper**. You may proceed with **QUESTION 1** first.

Shift
Lab

For Examiner's Use	
Section A	
1	/ 20
2	/20
3	/15
Total	/55

QUESTION 1

Investigation of glucose concentration in fruit juice.

You are advised to:

- *Read through the entire question and prepare a table to record your results in (b)(ii);*
- *Prepare a boiling water bath;*

before starting the investigation.

You are required to estimate the concentration of reducing sugar in a fruit juice, labelled **F1**, by comparison with that in a range of glucose solutions.

You are provided with a 0.8% solution of glucose, labelled **F2**, Benedict's solution, distilled water and five test-tubes.

(a) Carry out the Benedict's test on fruit juice **F1**. Describe the procedure. Record your results in your table in **(b)(ii)**.

.....[1]

1. Description of Benedict's test that works: add an appropriate **equal volume** (e.g. 2.0 cm³) of **sample** and **Benedict's solution** put in **boiling** water bath for appropriate duration (list a duration e.g. 10min)

(b) You are now going to test a range of glucose solutions that you will prepare yourself using **F2** and distilled water.

Carefully follow the instructions below. You should present and record your observations and data in the space provided. You will need to:

- read through the instructions carefully
- prepare the space on the next page so that it is ready for you to record the readings
- prepare a range of glucose of varying concentrations by **serial dilution**
- carry out the tests so that you can compare your results with the result for the fruit juice **F1**.
- rinse your syringes if necessary.

(i) You will now perform **serial dilution** using **F2** (0.80% glucose) as stock solution according to the steps below.

You will use **F2** (0.80% stock solution) to prepare 8.0 cm³ of glucose solution of **half** the concentration i.e. 0.40%.

Using this newly prepared 0.40% solution, prepare another 8.0 cm³ solution of **half** the concentration i.e. 0.20%.

Continue using this method until you have a total of 5 glucose solutions of varying concentrations.

Complete the table below.

concentration of glucose / %	0.80 (F2)	0.40	0.20	0.10	0.05
volume of the glucose solution to be diluted / cm ³		4.0	4.0	4.0	4.0
Volume of distilled water / cm ³		4.0	4.0	4.0	4.0

-[2]
1. Accuracy of volume to 1 d.p. for cm³ and concentration to 2 d.p.
 2. Correct concentrations of glucose AND appropriate volumes of glucose to be diluted and distilled water

(ii) Using a **table**, record the results for Benedict's test of F1 and the range of glucose standards in the space below.

-[3]
1. independent variable in leftmost column
 2. heading column headings include concentration with units in % and colour;
[trend of color going from brick red ppt, to orange, to yellow, to blue-green, to blue]

Concentration <u>of glucose</u> / %	Initial color <u>of Benedict's</u> <u>solution</u>	Final color <u>of Benedict's</u> <u>solution</u>
0.05	Blue solution	Blue solution /Green ppt in blue solution
0.10	Blue solution	Red ppt in blue solution
0.20	Blue solution	Reddish brown ppt
0.40	Blue solution	Orange ppt
0.80	Blue solution	Dark brown ppt
F1 (unknown)	Blue solution	Reddish brown ppt

(iii) Estimate the concentration of reducing sugar in the fruit juice **F1**. You may use a range for your estimation or the nearest glucose standard concentration.

- [1]
1. correct value/range for fruit juice concentration i.e. >0.10% and < 0.40%; Reject:
0.10%; 0.40% [ecf: dp]

(c) The volume of reactants can affect the results that you obtain.

(i) State how you controlled this variable in your investigation.

..... [1]
 1. Keep volume of **glucose** solutions measured constant for each test + list a **volume**
 e.g. 2.0cm^3

AND

keeping the volume of **Benedict's** solution constant for each test + list a **volume** e.g.
 2.0cm^3

AND

Listing of 10.0cm^3 syringe (**apparatus**)

(ii) Identify two other significant sources of error in this experiment. [2]

1. Difficulty in judging colour / colour identification is subjective
 2. Insufficient concentrations tested in glucose standards (thus, difficult to estimate F1 concentration accurately)
 3. Lack of replicates performed lead to lack of reliability / lack of repeats performed to ensure reproducibility of results
 4. Lack of (negative) control performed
- [Any 2]

(d) Suggest how the student could improve this experiment.

..... [3]
 [Difficulty in judging color]

1. use more accurate measuring device e.g. colorimeter/compare colour chart/spectrophotometer;

[Difficulty in estimating concentration of F1 using the 5 glucose standards]
 2. use wider range of solutions at different concentrations / prepare more concentrations of glucose

[Lack of replicates / repeats]

3. perform replicates/repeat at each concentration of glucose;
- [Lack of control]

4. replace glucose solution with equal volume of distilled water

(e) A student used another carbohydrate, starch, to investigate the effect of pH on the activity of the enzyme amylase.

The data in Table 1.1 were obtained.

Table 1.1

pH	time taken for complete hydrolysis / min			mean time / min
	first run	second run	third run	
5	9	10	8	9.0
6	7	6	8	7.0
7	3	4	3	3.3
8	4	5	6	5.0
9	10	9	11	10.0

(i) When the student first performed this investigation, the time taken for complete hydrolysis at pH 7 was 17 minutes.

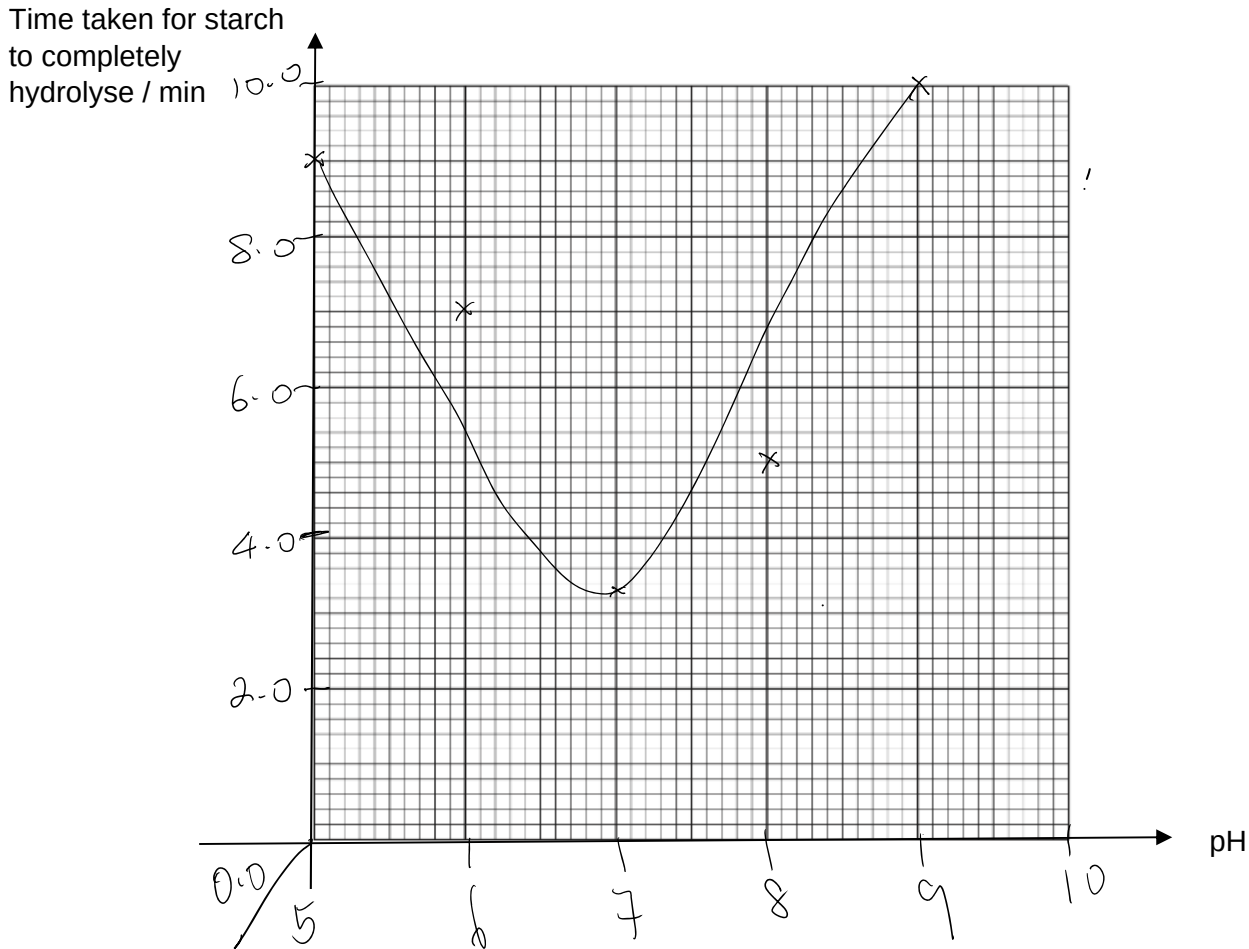
Explain why the student discarded this result and repeated the experiment with freshly made solutions.

.....[1]

1. time for that pH should be much quicker/AVP (accept: reading anomalous/not reliable)

(ii) Plot a graph to show the effect of pH on the time taken for complete hydrolysis of starch by amylase.

Graph showing effects of pH on time taken for starch to completely hydrolyse /min

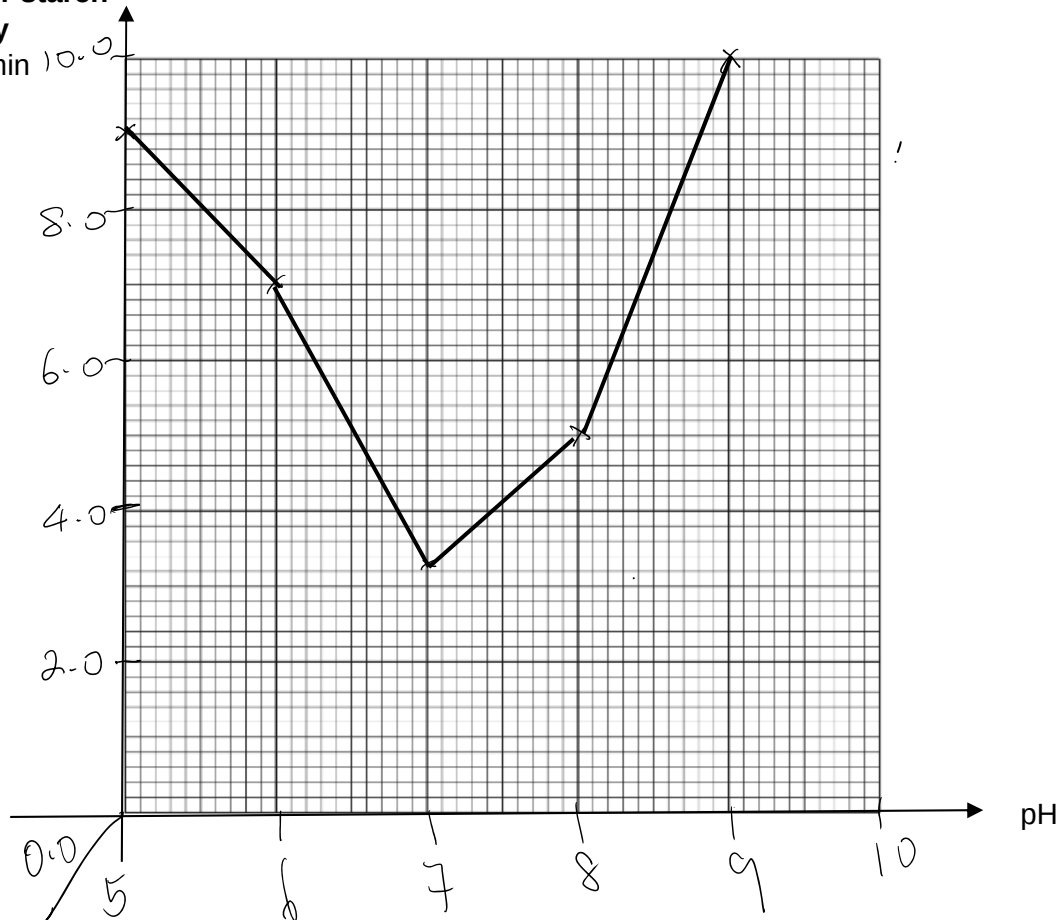


1. **[axes]** independent variable (pH) on x-axis, dependent variable (mean time/min) on y-axis AND axis **labels** appropriate (accept ecf from table if already penalised in (b) (i));
2. **[scale]** scale should be chosen so that data spans **at least half** of the width and height of the grid AND **scale appropriate** such as 1:10, 1:5 or 1:2 (Reject: awkward scales such as 3:10, 7:10, 8:10) (scale does not need to start at 0);
3. **[plot]** data plotted accurately to within 1mm, using crosses or circle-with-dot AND points joined with **straight ruled lines (dot to dot plot)** OR **fine curve (best fit line)** drawn through the data points, **not extrapolated** beyond the first or last point;

OR

Graph showing effects of pH on time taken for starch to completely hydrolyse /min

Time taken for starch to completely hydrolyse / min



1. **[axes]** independent variable (pH) on x-axis, dependent variable (mean time/min) on y-axis AND axis **labels** appropriate (accept ecf from table if already penalised in (b) (i));
2. **[scale]** scale should be chosen so that data spans **at least half** of the width and height of the grid AND **scale appropriate** such as 1:10, 1:5 or 1:2 (Reject: awkward scales such as 3:10, 7:10, 8:10) (scale does not need to start at 0);
3. **[plot]** data plotted accurately to within 1mm, using crosses or circle-with-dot AND points joined with **straight ruled lines (dot to dot plot)** OR **fine curve (best fit line)** drawn through the data points, **not extrapolated** beyond the first or last point;

(f) The student's hypothesis was:

- the activity of the enzyme would increase with increasing pH.

Discuss the student's hypothesis in relation to the results obtained.

.....[3]

1. Quote + Ref. results agree with hypothesis from **pH 5 to pH 7/optimum pH**
2. Quote + Ref. results do not agree with hypothesis from **pH 7/optimum pH to pH 9**
3. [compulsory] Ref. enzyme gradually denatures at **low** and **high** pH;

[Total : 20]

QUESTION 2

You will need access to a microscope and stage micrometer for this question.

Betalains are water-soluble red pigments found in some fungi and flowering plants, including rhubarb and beetroot.

The red colour of the rhubarb stem is a result of betalains, present in the cytoplasm of the rhubarb cells. This gives the cytoplasm a reddish colour, enabling the cell membrane to be distinguished.

You are required to investigate the effect of a solution, **labelled K**, on the epidermal cells of the rhubarb.

You are provided with a petri dish containing two segments of the rhubarb stem.

- Use a pair of forceps to carefully peel a piece of the epidermis (coloured) from one of the rhubarb stems.
- With the scalpel, cut a small piece (e.g. 0.5cm X 0.5cm) from this epidermal tissue.
- Mount the square on a microscope slide in **distilled water** and cover it with a cover slip.

You are also provided with an eyepiece graticule that has been fitted to the eyepiece of your microscope and a stage micrometer.

The 1cm stage micrometer is divided into 100 divisions.

(a) Observe your slide under the low-power and then high-power objective lens (40X) of your microscope.

(i) Use the space below to make a high-power drawing of **3** adjoining epidermal cells. Label your drawing.

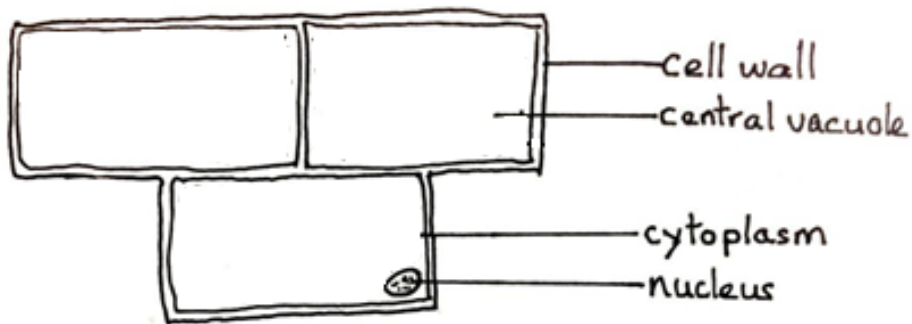
.....[5]

Mark scheme:

1. **3** adjoining cells drawn;
2. Cells drawn with clear, continuous lines;
3. **Labels:** Cell wall, plasma membrane/ cell **surface** membrane, cytoplasm,(nucleus);
4. Appropriate thickness of cell wall
5. Presence of **shared cell wall**

Reminder: draw inner layer separately, then outline the exterior for cell wall

High power drawing of rhubarb epidermal cells (400X)



- (ii) Count the number of divisions of the eyepiece graticule across any 1 of the 3 cells using the 40X objective lens.

Number of divisions

Range: 10 - 90 divisions

Remove the slide and replace it with the stage micrometer. Using the same magnification, adjust the focus until you can see the eyepiece graticule on top of the stage micrometer.

- (iii) Count the number of eyepiece graticule divisions that match an exact number of stage micrometer divisions.

Number of eyepiece graticule divisions

Ans: 40 graticule divisions

Number of stage micrometer divisions

Ans: 1

Use this information to calculate the actual diameter of the cell.

Show your working.

Actual diameter of cell [2]

Mark scheme:

1. Working shown (i.e. 10-90 divisions **X 2.5um**) [ecf: wrong number of divisions]
2. Actual diameter = 25um – 225um

- (iv) Suggest how an error in measuring the lengths could occur.

.....[1]

1. Thickness of scale lines / difficulty in matching the scales;

Reject: ref. any operator's error. i.e. parallax error in reading the number of divisions

- (b) Fig. 2.1 shows a photomicrograph of rhubarb cells after being immersed in **Solution K** for a few minutes.

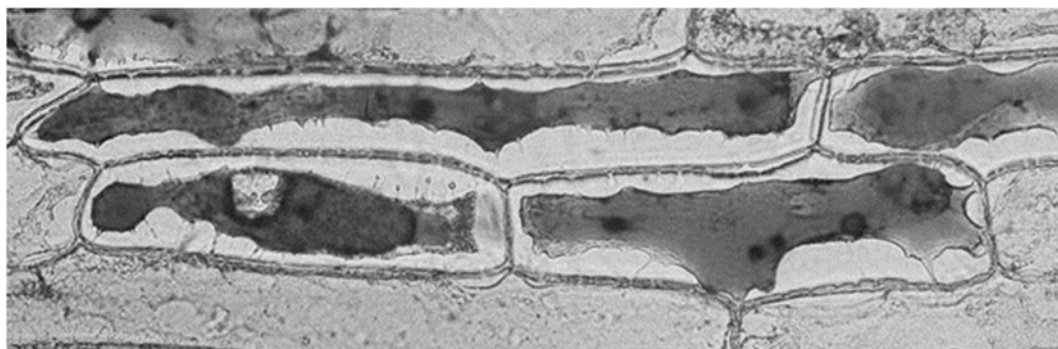


Fig. 2.1

- (i) Prepare the space below so that it is suitable for you to record the observable differences between the rhubarb cells you have seen in **2(a)** and the cells in **Fig. 2.1**.

Record these differences in the space you have prepared.

.....[2]

	Rhubarb cells in 2(a)	Rhubarb cells in Fig 2.1
1	No pulling away of cell surface membrane from cell wall;	Cell surface membrane pulls away from cell wall / shrinkage of cytoplasm;
2	No empty spaces within cell;	Empty spaces within the cell;
3	Cells are turgid/swollen;	Cells are flaccid/shrunken/not turgid/swollen;

NB: Must have table drawn with lines

Examiner's comments: If coloured diagram is given, can consider colour intensity as a difference (higher concentration of betalain pigments found in plasmolysed cells due to water moving out of cells)

Reject: vacuoles shrink after plasmolysis (as preamble in page 8 states that pigment is within cytoplasm)

- (ii) Explain the observation of the cells in **Fig. 2.1** and suggest a property of **Solution K**.

.....[4]

- Cells are **plasmolysed**;
- ref. movement of water out of cells by **osmosis** i.e. water leaving the rhubarb cells;
- from region of high water potential to region of lower water potential through a selectively/partially [Reject: semi] permeable / cell surface membrane
- [property of Solution K] Solution K is a hypertonic / concentrated solution

- (iii) With atmospheric temperature set to increase due to climate change, predict how the constituents of the plasma membranes of plants will change.

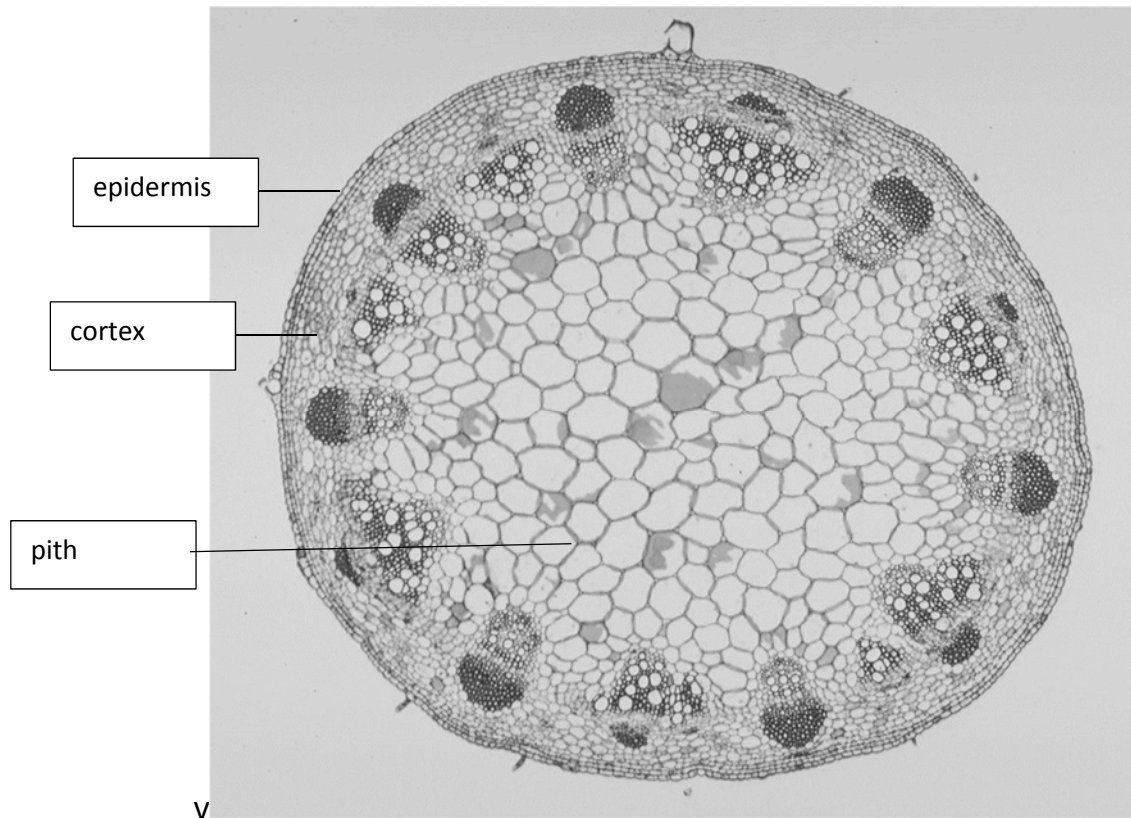
Explain why.

.....[3]

1. More cholesterol found in cell membrane;
2. Higher ratio of saturated fatty acids:unsaturated fatty acids
/ more saturated fatty acids / less unsaturated fatty acids / less C=C double bonds
3. ref. decrease fluidity of cell membrane / maintain integrity of membrane at a higher temperature;

(c) **Fig. 2.2** is a photomicrograph of a stained transverse section through an organ of a different type of plant.

You are not expected to be familiar with this specimen.



(Adapted from www.carolina.com)

Fig. 2.2

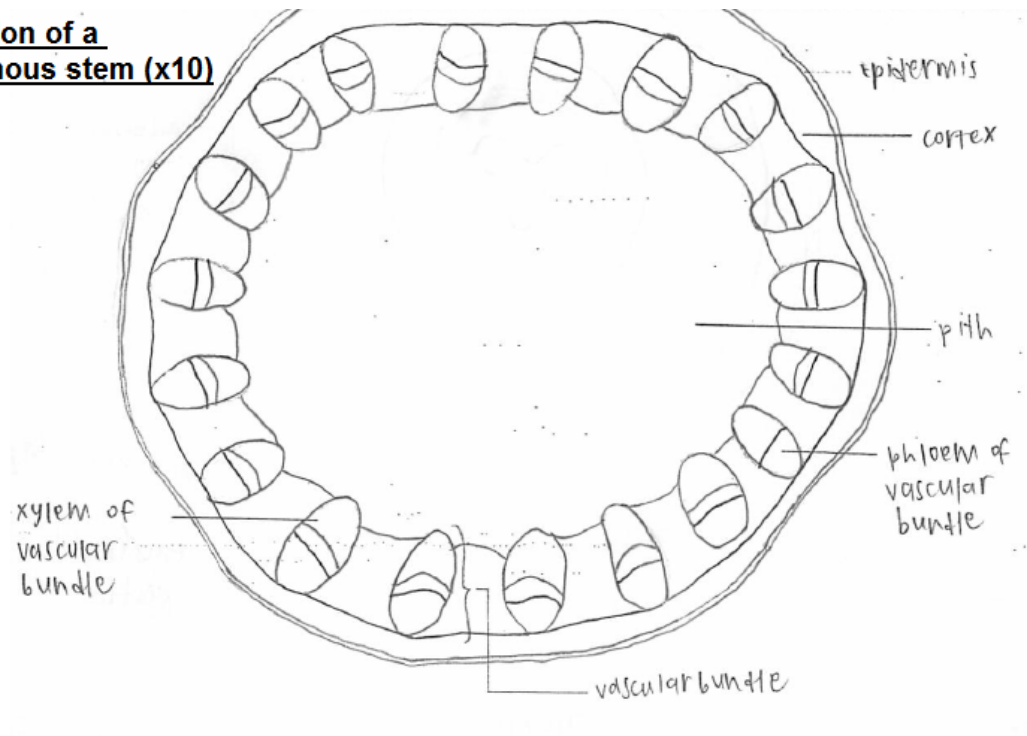
Make a low-power plan drawing of **Fig. 2.2**.

.....[3]

Mark scheme:

1. at least 3 layers shown, with appropriate thickness, e.g. epidermis, cortex, pith;
(ignore: layer joining the cambium)
2. at least 3 layers for vascular bundles e.g. xylem, phloem, cambium;
3. plan drawing with clear, continuous lines, no cells drawn and no shading;

**Cross section of a
dicotyledonous stem (x10)**



[TOTAL : 20]

QUESTION 3: PLANNING QUESTION

Effect of alcohol on membrane permeability of beetroot tissue.

Beetroot cells store molecules of betalain in their vacuoles. The membrane surrounding vacuoles in plant cells is the tonoplast, which has a similar structure to other cell membranes. Various factors such as alcohol influence the permeability of membranes.

Using this information and your own knowledge, design an experiment to determine the effect of different concentrations of alcohol on the membrane permeability of beetroot cells.

The following equipment may be used in your plan or not as you wish. You should **not** use any additional reagents/equipment.

- beetroot root tissue
- distilled water
- a supply of hot water
- Clingwrap
- 100% alcohol containing a mixture of ethanol and methanol
- thermometer
- thermostatically controlled water bath
- colourimeter & cuvettes
- stopwatch
- core borer
- ruler
- scalpel
- syringes
- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 15]

MARK SCHEME

INTRODUCTION

A: BACKGROUND KNOWLEDGE / RATIONALE

[1m for 2 points = 2m]

1. ref. plasma membrane / tonoplast membrane as a **phospholipid bilayer**, (with hydrophilic phosphate heads facing outwards and hydrophobic fatty acid tails facing inwards);
2. hydrophobic core / fatty acid tails are impermeable to charged/polar substances /allow non-polar substances to pass through;
3. Alcohol disrupts the physical structure of membranes / ref. denaturation of proteins embedded in membrane
4. ref. leakage of betalain molecules out of cytoplasm/vacuole/cell into surrounding medium

[Hypothesis] – 1m

5. As the alcohol concentration increases, the membrane permeability increases (followed by plateau)

[Rationale] – 1m

6. Membrane permeability is indicated by the extent of leakage of pigments which is **measured using a colourimeter to give an absorbance value.**

B: VARIABLES AND CONTROLLED VARIABLES

[State the independent and dependent variables] – 1m for #1,2

1. The independent variable is alcohol concentration / % ; 10%, 30%, 50%,70%, 90%. [At least 5 readings, regular intervals; maximum 100%]
2. The dependent variable is **membrane permeability**, measured by the absorbance of the surrounding medium.

[Other variables to keep constant: Can be written in detail in Procedure Section] – 1m for 2 points, max 1m

1. **Surface area/dimensions** and number of discs of beetroot tissue used
 - Use a core borer to ensure cylinders of beetroot tissue are of the same dimensions/size, and cut into discs of the same thickness e.g. **0.5cm using scalpel and ruler**
2. Volume of alcohol solution
 - Use a syringe to add the same volume e.g. 10cm³ (state volume here or in procedure) of fresh alcohol solution from the same stock (stirred well before use) to keep the initial concentration of alcohol constant.
3. Temperature
 - The temperature, **e.g. 35°C**, is kept constant by using a thermostatically controlled water bath/ thermometer and hot and cold water

4. pH

- Keep pH constant with a pH buffer of same volume (2cm³)

5. Duration of reactions

- A digital stopwatch is used to ensure duration of reactions is kept constant **e.g. 5 min.**

[Control] -1m

1. A control is set up with **10.0 cm³** of distilled water instead of alcohol solution, but with all other experimental conditions remaining constant.

Purpose: to show any change in absorbance of solution is due to the disruption of the membranes by alcohol.

2. A control is set up using the same dimensions/size/number of **plastic discs** (or other inert material), subjected to the same concentrations of alcohol and experimental conditions.

Purpose: To show that any change in absorbance of solution is due to the leakage of betalain pigments from the beetroot tissue.

[Diagram] -1m

Labelled with:

Beaker of water as conventional water bath with thermometer / thermostatically controlled water bath, test tube with discs

C: DETAILED PROCEDURE – total 7m**Part 1: Preparation of the different alcohol concentrations**

1. Label 5 boiling tubes – 10%, 30%, 50%, 70%, 90%. (minimum 5 tubes)
2. Prepare 30.0 cm³ of various concentrations of alcohol solutions as shown in the table below. 10.0 cm³ syringes are used to add the stock alcohol solution and distilled water into the respective boiling tubes.

1m

Concentration of alcohol solution to be prepared /%	Volume of 100% alcohol solution / cm ³	Volume of distilled water / cm ³
10	3.0	27.0
30	9.0	21.0
50	15.0	15.0
70	21.0	9.0
90	27.0	3.0

OR

Concentration of alcohol solution to be prepared /%	Volume of 100% alcohol solution / cm ³	Volume of distilled water / cm ³
20	4.0	16.0
40	8.0	12.0
60	12.0	8.0
80	16.0	4.0
100	20.0	0.0

Part 2:

3. Use a core borer to obtain 2-3 cylinders from the beetroot tissue.
4. Use a ruler and scalpel to measure and cut the cylinder into 5.0cm discs.
5. Wash the discs under running water to remove any pigments released from the cells during cutting.
6. Using a syringe, measure 10cm³ of 10% alcohol solution into a beaker.

1m

7. Cover the beaker with **Clingwrap to prevent evaporation** of alcohol.

1m

8. Incubate the beaker containing the alcohol solution in a thermostatically controlled water bath at **35°C. Allow 5 minutes** for the alcohol solution to **equilibrate**.
9. Place 2 beetroot discs into the beaker containing 10% alcohol solution.
10. Ensure that the discs are completely soaked in the alcohol solution.
11. Using a stopwatch, incubate the beaker containing the beetroot discs in the thermostatically controlled water bath for 15 minutes. Ensure that beaker is covered to prevent evaporation of alcohol.
12. After 15 minutes, remove the beetroot discs from the solution.

13. Pour 2.0cm³ of the alcohol solution into a cuvette. Shake the solution gently to ensure that the pigment is well mixed into the water before pouring into the cuvette.

1m

14. Measure the absorbance of the solution using a colourimeter. Record the absorbance into a suitable table.

15. Repeat steps 5 to 11 using the different concentrations of alcohol solutions.

[Replicates and Repeats] – 1m

16. To ensure reliability of results, repeat steps 3 to 14 to obtain a total of three readings at each alcohol concentration, and calculate the average.

17. To ensure reproducibility of data, repeat the entire experiment twice using freshly prepared reagents and solutions and beetroot root tissue.

D: DATA MANIPULATION AND EXPECTED RESULTS

1. [Draw Table of results] -1m

Table showing the absorbance /A at different concentrations of alcohol /%

For control using 0% alcohol

Concentration of alcohol /%	Absorbance of solution /A			
	Reading 1	Reading 2	Reading 3	Average
0				
10				
30				
50				
70				
90				

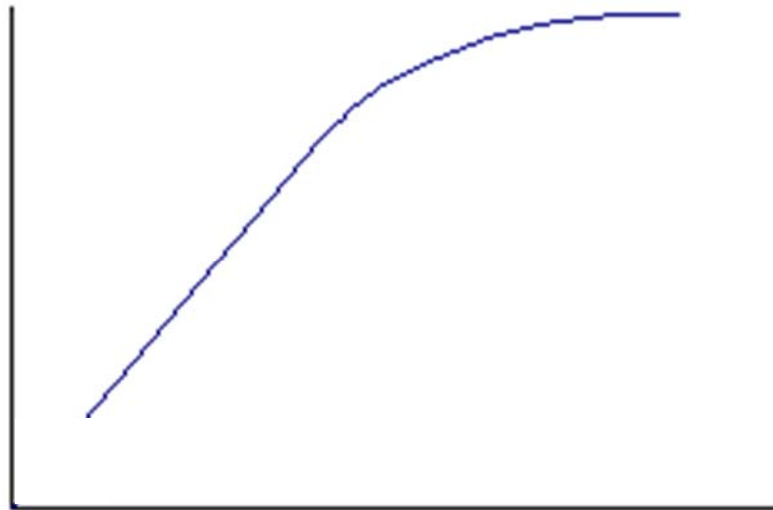
For control using inert plastic discs:

Concentration of alcohol /%	Absorbance of solution /A						
	Reading 1	Control	Reading 2	Control	Reading 3	Control	Average
0							
10							
30							
50							
70							
90							

2. [Draw a graph to show expected trends and/or results] -1m

Graph of absorbance /A against concentration of alcohol /%

Membrane permeability or Absorbance /a.u.



Concentration
of alcohol /%

SAFETY PRECAUTIONS -1m for 2 sets

	Risk	Precaution
1	Alcohol is flammable.	Ensure that there is no naked/open flame nearby.
2a	Scalpel is sharp and cause injuries	Place the sharp objects away from the main work area after use / handle with caution
2b	Core borer is sharp and cause cuts	Handle with caution
3	Alcohol is an irritant/toxic.	Wear safety goggles and gloves when handling alcohol.
4	AVP	

[Total: 15]